

Online Appendix: [From Experts to Language Models: Enhancing Online Expert Elicitation with LLMs]

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1 Document Preamble

This document is identified as the **Online Appendix**. It contains the figures and tables referenced in the paper.

2 The DC Solutions

This part provides information on the Direct Current (DC) solutions, including their application sites, names, and respective descriptions.

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Table A1. Descriptions of DC Solutions

S/No	Demonstrators	DC Solution	Description
1	All demonstrators	Smart and sustainable DC cables	These are designed with durable, environmentally friendly materials, ensuring comparable or improved longevity relative to AC cables. The cables prioritize ergonomic use, mimicking standard AC cable handling to simplify adoption across all demonstration sites.
2	All demonstrators	DC connectors	The project is developing two types of connectors—passive and smart. Passive connectors offer enhanced physical robustness, while smart connectors incorporate microelectronics to handle higher voltages and currents.
3	Buildings	Static protection system	This solution includes advanced fault detection and ultra-fast protection devices that can rapidly identify and isolate DC faults within microgrids.
4	Industry and buildings	Semiconductor-based circuit breaker	Designed to overcome the absence of natural current zero-crossing in DC systems, this breaker ensures rapid fault interruption. It complies with Current/OS and ODCA standards and incorporates adaptive protection schemes, real-time monitoring, and communication with central controllers.
5	All demonstrators	Protection DC system design tool	A software tool that facilitates the design and simulation of protection systems for DC grids.
6	Buildings	DC-DC converter	Enables power flow control among DC appliances. The converter features real-time monitoring of voltage, current, and temperature, as well as load balancing and peak shaving functionalities for improved grid performance.
7	Ports, industry, and buildings	LVAC-LVDC inter-link converter	A low-voltage AC-DC converter designed to bridge AC and DC systems. It includes built-in droop control to reduce the number of downstream devices needed in the DC architecture.
8	All demonstrators	DC measurement device	Developed by XXXXXX XXXXXX, this solution enables easy retrofit installation and accurate, reproducible measurements without requiring on-site calibration. It aggregates multiple sensors into a single robust measurement system.
9	All demonstrators	DC solution design tool	A user-friendly, possibly open-source software tool supporting the design of DC systems. It accommodates models for technologies in the XXXXXX project and aligns with both Current-OS and ODCA frameworks. The tool emphasizes drag-and-drop usability and supports electrical sizing and techno-economic analysis.
10	All demonstrators	Network design tool for DC solutions	A simulation tool that integrates static models of key DC devices and supports multiple DC ecosystem configurations.
11	Industry	Solid-state circuit breaker	A high-speed circuit breaker capable of fault detection and isolation in microseconds. It emphasizes thermal and current handling realism and includes real-time monitoring and IoT-enabled communication features

3 Prompts Information

This section describes the components that formed the prompts used for the Large Language Model (LLM) variants in the zero-shot, context, and anchored experiments.

Table A2. Descriptions used for each experiment—zero-shot, context, & anchor

S/No	Instruction Type	Instruction Text	Experiment Used	Question Type
1	Description	– The expert elicitation is based on the XXXXX project, which aims to accelerate the transition to a sustainable energy system by promoting the use of direct current (DC) technologies. – The project focuses on the development and demonstration of DC solutions in four sectors: ports, industry, data centers, and buildings.	Zero-shot	All questions
2	DC solutions	List of all 11 DC solutions	Zero-shot	Feasibility & importance
3	Project descriptions	– The project is part of the XXXXX initiative. It focuses on establishing guidelines and roadmaps to enable the widespread adoption of DC solutions, especially medium- and low-voltage DC. The project will develop multiple technologies, including software tools, simulation tools, and hardware solutions such as cables and converters. – Because the project is still early, it has not yet engaged end users extensively. Instead, it examines topics reported in the literature by gathering input from experts in DC systems. Experts are drawn from several areas, including ports, buildings, manufacturing, and utilities, all aligned with the project’s energy and DC expertise. – The project is organized around: • An overview of DC grids • Development of tools and IT solution architectures (selected as enablers) • Prototypes (including items with high TRLs) • Simulation tools • Four demonstrations across application areas selected by the commission: data centers, buildings, industry, and ports – The project also includes feasibility studies and recommendations, culminating in DC roadmaps based on the results.	Context & anchor	All questions

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Table A2 – Continued from previous page

S/No	Instruction Type	Instruction Text	Experiment Used	Question Type
4	Demonstrators descriptions	Four demonstrators are planned across data centers, buildings, industry, and ports, described as follows: <i>Data center:</i> A data center demonstrator is planned in XXXXX. The focus is on edge data centers and how data centers can be integrated, particularly with renewables and heat reuse. The demonstrator also considers powering the office via DC, not only the computing equipment. A micro data center will be tested in XXXXX with a small installation assembled from data center components. The installation includes vehicle-to-grid (V2G) and solar panels, and it will operate as a real micro data center. <i>Building:</i> A residential building demonstrator in XXXXX will test how existing DC technologies can be integrated into a DC grid. The buildings demonstrator includes a live setup and focuses on DC-related technologies, including distributed photovoltaic (DPV) systems. <i>Industry:</i> Two industry use cases are planned. One manufacturing demonstrator is located in XXXXX, XXXXX, where technologies will be integrated and large facilities will be connected. Another industry demonstrator is in XXXXX Contact, focusing on accommodating renewables and chargers, with a scale around 100 kW. A separate industry/living-lab setup in XXXXX connects two buildings using DC technology. <i>Port:</i> A port demonstrator is planned in XXXXX. The objective is to assess the feasibility of DC for onshore power supply in port environments. The port demonstrator is small and includes a digital twin to explore scaling of DC in the port. In addition, the demonstrator tests multiple technologies and considers end-user perspectives, including people passing by (e.g., tourists) who may not realize the system is DC. The installation includes approximately 10 kW of PV and 11 kW of V2G chargers.	Context & anchor	All questions

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Table A2 – Continued from previous page

S/No	Instruction Type	Instruction Text	Experiment Used	Question Type
5	Elicitation instructions	– The elicitation process is part of the XXXXX XXXXX project and focuses on users’ perceptions of medium- and low-voltage DC technologies. – Because the project is still in its first year and there are few direct current users at present, the study uses experts instead of relying on user data. – The goal is to gain insights into the challenges and opportunities for medium- and low-voltage DC, including areas where there is still uncertainty and mixed views in the literature. – The protocol is designed to gather qualitative and quantitative input from experts through a set of prepared questions. – The elicitation will be used to document expert opinions, and to converge on areas of agreement and identify areas of disagreement. – The session includes brief introductions, a short overview of the broader project kept general rather than limited to the XXXXX project, an explanation of the expert elicitation protocol and its process, and concluding remarks. – Topics selected for discussion include barriers to adoption, costs, time, risks of delay, and related general considerations.	Context & anchor	All questions
6	DC solutions	List of all 11 DC solutions with their respective description on each, as provided in Table A1	Context & anchor	Feasibility & importance
7	Barriers	List of barriers as provided in the paper (see Table 2)	Zero-shot, context, and anchor	Barrier-select

Table A3. BERT-score semantic fidelity assessment for the initial runs and the final revision.

S/No	Raw file	Edited file	Precision	Recall	F1	Status
1	Section: elicitation					
	transcription	cleaned_contexts_run01_20260629_104226	0.820	0.823	0.822	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.827	0.820	0.824	not retained
	transcription	cleaned_contexts_run02_20260629_104239	0.839	0.819	0.829	retained
	author-review-raw	author-review-edited	0.976	0.936	0.956	final version
2.	Section: project					
	transcription	cleaned_contexts_run01_20260629_104226	0.862	0.842	0.852	not retained
	transcription	cleaned_contexts_run02_20260629_104239	0.860	0.846	0.853	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.870	0.845	0.857	retained
	author-review-raw	author-review-edited	0.983	0.968	0.975	final version
3.	Section: demonstrator					
	transcription	cleaned_contexts_run02_20260629_104239	0.809	0.842	0.825	not retained
	transcription	cleaned_contexts_run03_20260629_104252	0.834	0.858	0.846	not retained
	transcription	cleaned_contexts_run01_20260629_104226	0.869	0.856	0.862	retained
	author-review-raw	author-review-edited	0.952	0.948	0.950	final version

4 Context Preparations

This section presents the results of the two BERTScore-based semantic fidelity assessments of the common content used in the context and anchor experiments. Semantic fidelity was assessed at two levels: between the raw content and the GPT-5.4-nano-generated cleaned content, and between the GPT-generated content and the author-cleaned content, with the author-cleaned version retained as final.

5 Anchor Phrases

This section contains the literature sources for the anchor phrases. It maps the literature sources to the phrases used in the anchor experiments for the feasibility, importance, and barrier-selection questions.

Table A4. Sources used to support each anchor phrase in the anchor framing prompts.

S/No	Anchor type	Anchor phrase	Year	Author/Org.	Citation
1	WORD	European deployments and standardization efforts indicate workable LV/MV-DC microgrids across sectors.	-	-	Derived from other anchor sources
2	EXAMPLE	DC microgrids integrating PV, storage, and DC loads (e.g., ± 380 V DC bus) are reported in operational testbeds.	2020	Fraunhofer IISB	[4]
			2014	TU Eindhoven / Vicor	[9]
			2019	EE Power	[5]
			2023	ETSI	[2]
3	NUM_LOW	Some peer-reviewed studies report 2–6% distribution-efficiency improvement with DC microgrids under efficient baselines.	2025	MDPI Energies	[1]
			2015	Aalborg University	[7]
4	NUM_HIGH	Other analyses report 10–18% distribution-efficiency improvement in modeled DC microgrid scenarios with PV and storage.	2023	MDPI Energies	[6]
			2020	Fraunhofer	[3]
			2024	SETIS / EU	[8]

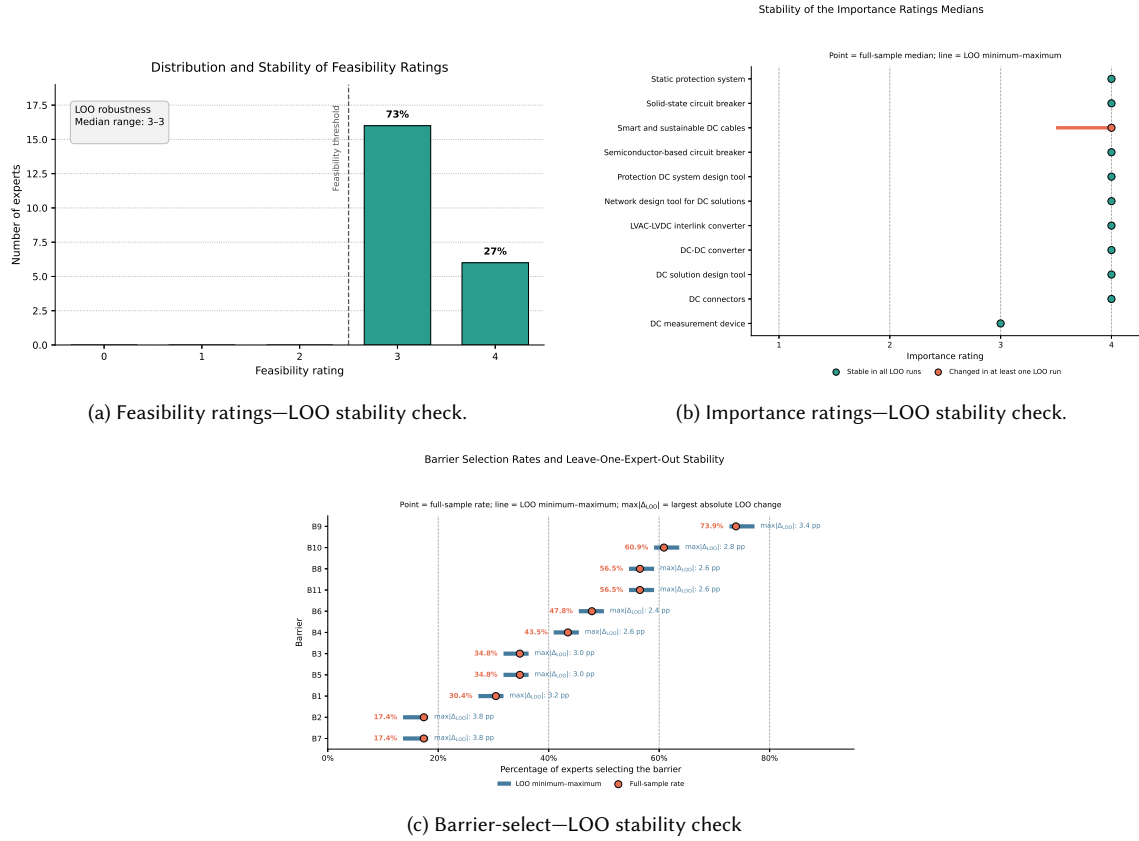
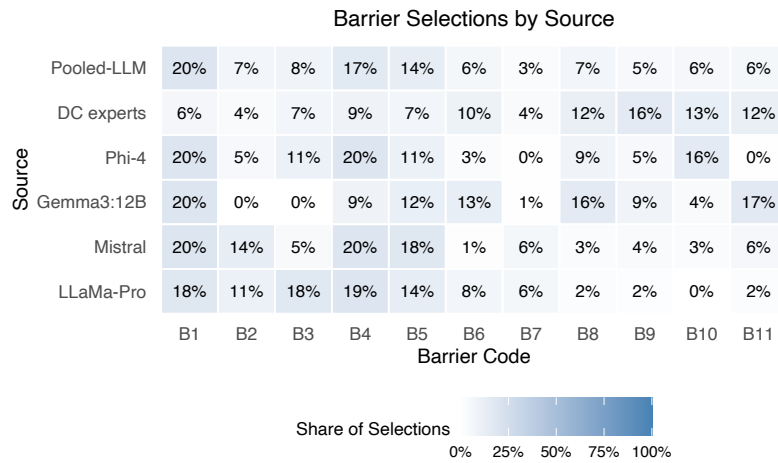


Fig. A1. Stability check on experts' responses—feasibility, importance, and barrier-select

6 Stability Evaluation of DC Experts' Responses—RQ1

This section presents the results of the stability check on experts' responses to the three questions: feasibility, importance, and barrier selection.



Note: Percentage in the heatmap shows the selection rate per barrier across the compared sources (i.e., DC experts and models). Whereby total selection per source sums to 100%.

Fig. A2. Barrier selection—models vs. DC experts

7 Barrier Selection Rate—DC Experts vs. Models

This section contains the barrier selection rates for the 11 barriers across DC experts, pooled LLMs, and each LLM individually; as supporting evidence for *RQ1*.



Fig. A3. Predictive posterior checks-RQ2 feasibility

Table A5. RQ2 feasibility: context vs. zero-shot prior sensitivity comparisons.

Model	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Phi-4	0.220 [0.097, 0.473]	0.139 [0.047, 0.346]	0.117 [0.037, 0.312]	Lower in Context / Lower in Context / Lower in Context	Yes
Gemma3:12B	1.232 [0.391, 3.908]	1.502 [0.292, 8.297]	1.684 [0.290, 11.686]	Uncertain / Uncertain / Uncertain	Yes
Mistral	1.238 [0.384, 3.997]	1.504 [0.306, 8.542]	1.698 [0.275, 12.396]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	2.979 [1.883, 4.738]	3.254 [2.026, 5.337]	3.329 [2.050, 5.461]	Higher in Context / Higher in Context / Higher in Context	Yes

Note. Cells report posterior median odds ratios with 95% credible intervals. R/P/W denotes regularizing, primary, and weak priors, respectively. Stable indicates that the inferential conclusion was identical under all three priors. OR > 1 indicates higher odds of a higher feasibility rating in context relative to zero-shot.

8 Posterior Predictive Check, and Sensitivity Comparisons—RQ2

8.1 RQ2 Feasibility Question

This part presents the results of the Predictive Posterior Check (PPC) and the sensitivity and robustness checks for the feasibility question of RQ2.

8.2 RQ2 importance question

This section contains the long table of the primary results, the PPC, and the sensitivity check for the importance question in RQ2. However, the sensitivity check is limited to contrasts of LLMs for the DC solutions whose conclusions were found to be sensitive in the full run.

Table A6. RQ2 importance: posterior effects of context relative to zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\text{OR} > 1)$	Interpretation
Gemma3:12B	DC connectors	0.003	[0.001, 0.007]	< 0.001	Lower in Context
Gemma3:12B	DC measurement device	0.019	[0.007, 0.043]	< 0.001	Lower in Context
Gemma3:12B	DC solution design tool	3.916	[0.552, 36.900]	0.903	Uncertain
Gemma3:12B	DC-DC converter	0.001	$[4.71 \times 10^{-4}, 0.004]$	< 0.001	Lower in Context
Gemma3:12B	LVAC-LVDC interlink converter	0.014	[0.005, 0.034]	< 0.001	Lower in Context
Gemma3:12B	Network design tool for DC solutions	1.129	[0.688, 1.874]	0.675	Uncertain
Gemma3:12B	Protection DC system design tool	106.152	[33.825, 490.424]	> 0.999	Higher in Context
Gemma3:12B	Semiconductor-based circuit breaker	3.967	[0.539, 37.652]	0.912	Uncertain
Gemma3:12B	Smart and sustainable DC cables	23.185	[5.659, 147.700]	> 0.999	Higher in Context
Gemma3:12B	Solid-state circuit breaker	3.862	[0.530, 41.682]	0.910	Uncertain
Gemma3:12B	Static protection system	1.85×10^3	$[537.585, 8.12 \times 10^3]$	> 0.999	Higher in Context
LLaMa-Pro	DC connectors	9.912	[4.860, 22.219]	> 0.999	Higher in Context
LLaMa-Pro	DC measurement device	0.863	[0.408, 1.747]	0.342	Uncertain
LLaMa-Pro	DC solution design tool	2.544	[1.470, 4.562]	> 0.999	Higher in Context
LLaMa-Pro	DC-DC converter	1.714	[1.110, 2.693]	0.990	Higher in Context
LLaMa-Pro	LVAC-LVDC interlink converter	0.638	[0.360, 1.117]	0.059	Uncertain
LLaMa-Pro	Network design tool for DC solutions	1.709	[1.048, 2.775]	0.987	Higher in Context
LLaMa-Pro	Protection DC system design tool	1.670	[1.048, 2.692]	0.984	Higher in Context
LLaMa-Pro	Semiconductor-based circuit breaker	113.719	[42.297, 398.501]	> 0.999	Higher in Context

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Table A6. RQ2 importance: posterior effects of Context relative to Zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\text{OR} > 1)$	Interpretation
LLaMa-Pro	Smart and sustainable DC cables	1.409	[0.893, 2.228]	0.932	Uncertain
LLaMa-Pro	Solid-state circuit breaker	47.547	[22.057, 117.553]	> 0.999	Higher in Context
LLaMa-Pro	Static protection system	41.511	[18.664, 109.765]	> 0.999	Higher in Context
Mistral	DC connectors	0.558	[0.142, 2.152]	0.197	Uncertain
Mistral	DC measurement device	0.126	[0.072, 0.211]	< 0.001	Lower in Context
Mistral	DC solution design tool	5.515	[1.429, 30.078]	0.995	Higher in Context
Mistral	DC-DC converter	0.106	[0.047, 0.218]	< 0.001	Lower in Context
Mistral	LVAC-LVDC interlink converter	1.229	[0.773, 1.971]	0.808	Uncertain
Mistral	Network design tool for DC solutions	3.893	[0.533, 40.869]	0.904	Uncertain
Mistral	Protection DC system design tool	9.777	[5.689, 17.474]	> 0.999	Higher in Context
Mistral	Semiconductor-based circuit breaker	205.628	[83.133, 600.933]	> 0.999	Higher in Context
Mistral	Smart and sustainable DC cables	0.126	[0.070, 0.214]	< 0.001	Lower in Context
Mistral	Solid-state circuit breaker	137.006	[44.153, 614.028]	> 0.999	Higher in Context
Mistral	Static protection system	917.910	[297.154, 3.47×10^3]	> 0.999	Higher in Context
Phi-4	DC connectors	0.310	[0.133, 0.662]	0.002	Lower in Context
Phi-4	DC measurement device	0.586	[0.368, 0.924]	0.013	Lower in Context
Phi-4	DC solution design tool	7.443	[1.440, 61.240]	0.993	Higher in Context
Phi-4	DC-DC converter	3.890	[0.544, 38.904]	0.909	Uncertain
Phi-4	LVAC-LVDC interlink converter	7.033	[4.210, 12.203]	> 0.999	Higher in Context
Phi-4	Network design tool for DC solutions	10.745	[2.250, 81.151]	> 0.999	Higher in Context
Phi-4	Protection DC system design tool	10.763	[2.381, 73.933]	> 0.999	Higher in Context
Phi-4	Semiconductor-based circuit breaker	11.671	[2.697, 78.429]	> 0.999	Higher in Context

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Table A7. RQ2 importance posterior predictive checks

Model	Within 95% PPI	Coverage	Median absolute difference	Maximum absolute difference
Phi-4	66/66	100.0%	0.007	0.013
Gemma3:12B	59/66	89.4%	0.007	0.053
Mistral	66/66	100.0%	0.007	0.033
LLaMa-Pro	66/66	100.0%	0.005	0.020
Overall	257/264	97.3%	0.007	0.053

Note. PPI denotes the 95% posterior predictive interval. Coverage is the proportion of observed rating-category proportions that fell within their corresponding PPIs. Absolute differences compare the observed rating-category proportions with the posterior-predicted median proportions. Each base model contributed 66 checks: 11 DC solutions $\times$ 2 conditions $\times$ 3 observed rating categories. The overall row contains all 264 checks.

Table A8. RQ2 importance contrasts whose conclusions were sensitive to the prior specification.

Model	DC solution	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)
Phi-4	DC solution design tool	2.688 [0.916, 8.645]; Uncertain	7.443 [1.440, 61.240]; Higher in Context	19.242 [2.065, 622.900]; Higher in Context
Phi-4	Solid-state circuit breaker	1.980 [0.570, 7.026]; Uncertain	4.701 [0.751, 43.782]; Uncertain	12.196 [1.115, 456.846]; Higher in Context

Note. Cells report the posterior median OR, 95% credible interval, and inferential conclusion under each prior. Only contrasts for which either direction or conclusion changed across priors are shown.

Table A6. RQ2 importance: posterior effects of Context relative to Zero-shot for each base model and DC solution under the primary prior.

Model	DC solution	OR	95% CrI	$P(\text{OR} > 1)$	Interpretation
Phi-4	Smart and sustainable DC cables	3.854	[0.540, 40.953]	0.909	Uncertain
Phi-4	Solid-state circuit breaker	4.701	[0.751, 43.782]	0.948	Uncertain
Phi-4	Static protection system	39.857	[11.877, 219.852]	> 0.999	Higher in Context

Note. Each row represents a separately estimated base-model/DC-solution contrast; the odds ratios were not pooled across DC solutions. $\text{OR} > 1$ indicates higher odds of receiving a higher importance rating in context relative to zero-shot, whereas $\text{OR} < 1$ indicates lower odds in context. The CrI denotes the 95% Bayesian credible interval. $P(\text{OR} > 1)$ denotes the posterior probability that the odds ratio exceeds one.

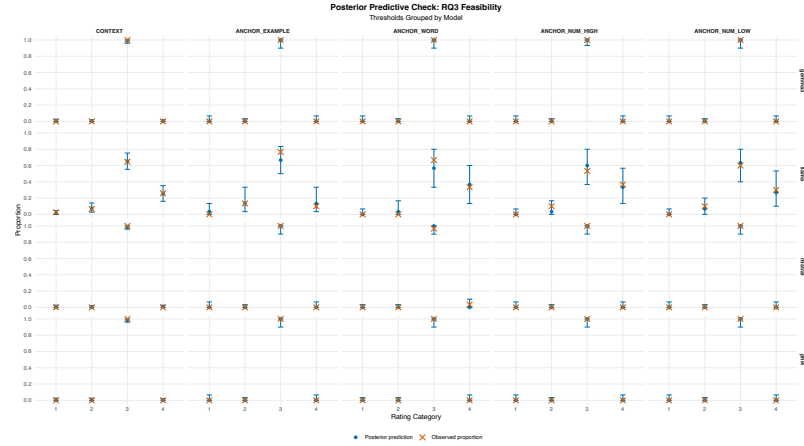


Fig. A4. RQ3 feasibility: posterior predictive checks

9 Posterior Predictive Check, Sensitivity Comparisons, and Anchor Patterns—RQ3

This section contains the PPC and sensitivity checks results from both feasibility and importance questions for RQ3.

9.0.1 RQ3 feasibility question. This part presents results in two main parts: the PPC and the sensitivity checks for feasibility ratings in RQ3.

Table A9. RQ3 feasibility effects of each anchor type relative to context prior-sensitivity comparisons

Model	Anchor type	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Phi-4	Example	1.091 [0.284, 3.934]	1.210 [0.156, 10.258]	1.450 [0.110, 24.053]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	Word	1.067 [0.292, 4.122]	1.218 [0.160, 10.844]	1.455 [0.120, 21.459]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	High numeric	1.067 [0.292, 4.074]	1.208 [0.166, 10.510]	1.457 [0.115, 21.753]	Uncertain / Uncertain / Uncertain	Yes
Phi-4	Low numeric	1.052 [0.280, 4.040]	1.228 [0.151, 10.851]	1.435 [0.123, 22.042]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	Example	1.077 [0.283, 4.131]	1.227 [0.162, 10.743]	1.443 [0.116, 20.737]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	Word	1.076 [0.287, 4.059]	1.226 [0.155, 10.752]	1.501 [0.115, 21.662]	Uncertain / Uncertain / Uncertain	Yes
Gemma3:12B	High numeric	1.069 [0.287, 4.011]	1.257 [0.163, 10.513]	1.462 [0.114, 20.139]	Uncertain / Uncertain / Uncertain	Yes

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Table A9. Prior-sensitivity comparison for the RQ3 feasibility effects of each anchor type relative to Context.

Model	Anchor type	Regularizing OR (95% CrI)	Primary OR (95% CrI)	Weak OR (95% CrI)	Conclusions (R/P/W)	Stable
Gemma3:12B	Low numeric	1.071 [0.299, 3.887]	1.260 [0.168, 11.360]	1.403 [0.121, 21.725]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Example	1.038 [0.281, 3.766]	1.145 [0.151, 9.070]	1.324 [0.114, 17.059]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Word	1.574 [0.441, 5.722]	3.084 [0.404, 21.888]	5.407 [0.414, 56.770]	Uncertain / Uncertain / Uncertain	Yes
Mistral	High numeric	1.036 [0.291, 3.763]	1.168 [0.160, 9.160]	1.350 [0.120, 18.329]	Uncertain / Uncertain / Uncertain	Yes
Mistral	Low numeric	1.011 [0.268, 3.825]	1.164 [0.156, 8.832]	1.336 [0.116, 19.036]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Example	0.569 [0.276, 1.157]	0.525 [0.242, 1.118]	0.506 [0.228, 1.117]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Word	1.563 [0.790, 3.056]	1.696 [0.795, 3.658]	1.726 [0.809, 3.678]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	High numeric	1.420 [0.705, 2.828]	1.509 [0.685, 3.266]	1.524 [0.695, 3.370]	Uncertain / Uncertain / Uncertain	Yes
LLaMa-Pro	Low numeric	1.138 [0.564, 2.231]	1.189 [0.549, 2.523]	1.186 [0.523, 2.647]	Uncertain / Uncertain / Uncertain	Yes

Note. Cells report posterior median odds ratios with 95% credible intervals. R/P/W denotes regularizing, primary, and weak priors, respectively. Stable indicates that the inferential conclusion was identical under all three priors. OR > 1 indicates higher odds of a higher feasibility rating under the specified anchor relative to Context.

9.1 RQ3 importance question

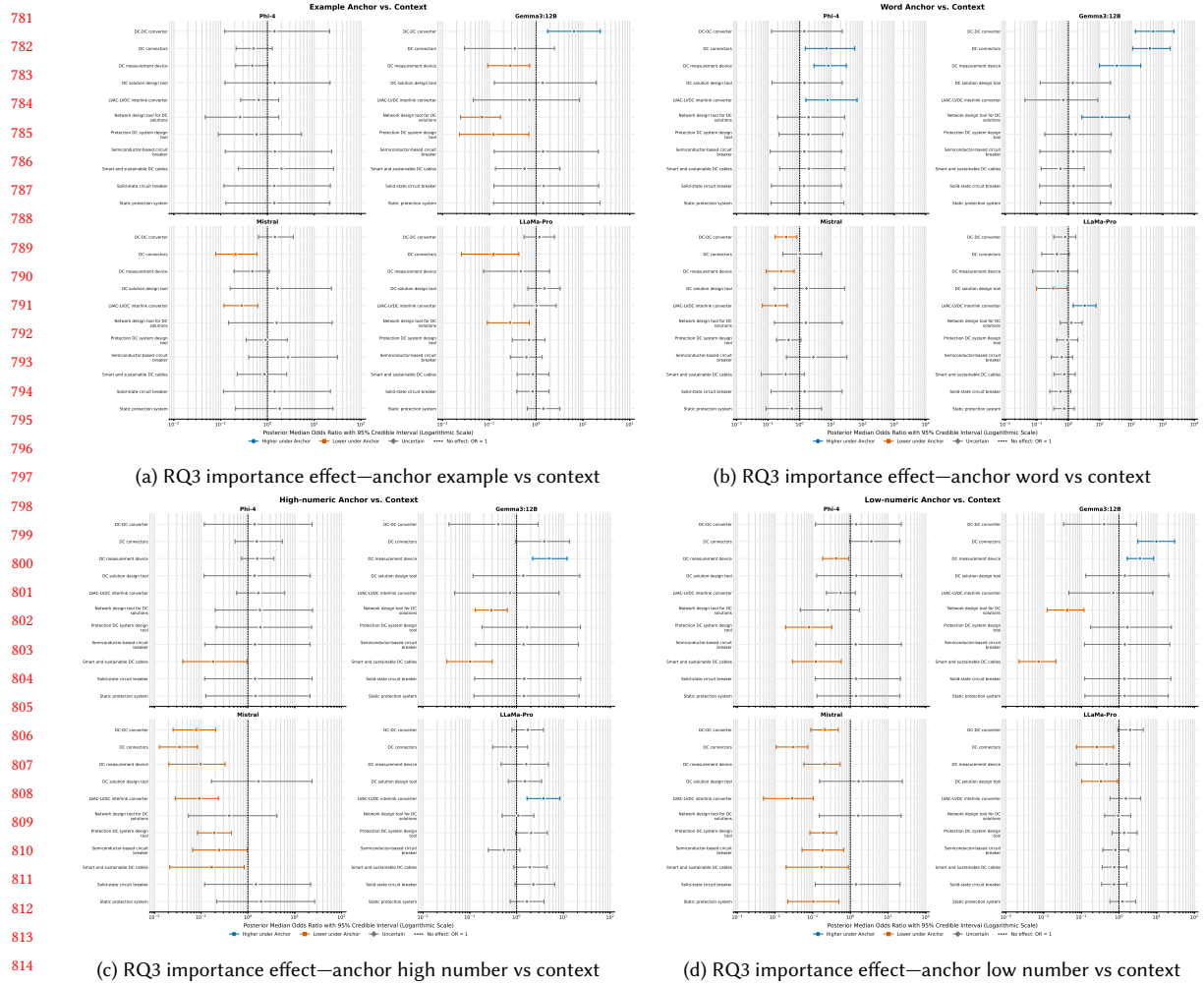


Fig. A5. RQ3 importance effect on primary priors for each anchor vs context

Table A10. RQ3 importance: primary model posterior predictive checks

Model	Comparison group	Within 95% PPI	Coverage	Med.abs dif- ference	Max.abs dif- ference
Phi-4	Context	22/22	100.0%	0.007	0.013
Phi-4	Example anchor	22/22	100.0%	0.000	0.033
Phi-4	Word anchor	22/22	100.0%	0.000	0.033
Phi-4	High-numeric anchor	22/22	100.0%	0.000	0.033
Phi-4	Low-numeric anchor	22/22	100.0%	0.000	0.067
Gemma3:12B	Context	22/22	100.0%	0.007	0.027
Gemma3:12B	Example anchor	22/22	100.0%	0.000	0.033
Gemma3:12B	Word anchor	22/22	100.0%	0.000	0.067
Gemma3:12B	High-numeric anchor	22/22	100.0%	0.000	0.033
Gemma3:12B	Low-numeric anchor	22/22	100.0%	0.000	0.067
Mistral	Context	22/22	100.0%	0.013	0.027
Mistral	Example anchor	22/22	100.0%	0.000	0.033
Mistral	Word anchor	22/22	100.0%	0.000	0.033
Mistral	High-numeric anchor	22/22	100.0%	0.033	0.033
Mistral	Low-numeric anchor	22/22	100.0%	0.033	0.033
LLaMa-Pro	Context	22/22	100.0%	0.003	0.013
LLaMa-Pro	Example anchor	22/22	100.0%	0.000	0.033
LLaMa-Pro	Word anchor	22/22	100.0%	0.000	0.017
LLaMa-Pro	High-numeric anchor	22/22	100.0%	0.000	0.033
LLaMa-Pro	Low-numeric anchor	22/22	100.0%	0.000	0.000
Overall	Overall	440/440	100.0%	0.000	0.067

Note. PPI denotes the 95% posterior predictive interval. Coverage is the proportion of observed rating-category proportions within their corresponding PPIs. Absolute differences compare observed proportions with posterior-predicted median proportions.

Table A11. Prior-sensitivity summary for the RQ3 importance anchor effects.

Model	Anchor type	Contrasts	Direction stable	Conclusion stable
Phi-4	Example	11	11/11 (100.0%)	11/11 (100.0%)
Phi-4	Word	11	11/11 (100.0%)	10/11 (90.9%)
Phi-4	High numeric	11	11/11 (100.0%)	10/11 (90.9%)
Phi-4	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
Gemma3:12B	Example	11	11/11 (100.0%)	9/11 (81.8%)
Gemma3:12B	Word	11	11/11 (100.0%)	11/11 (100.0%)
Gemma3:12B	High numeric	11	11/11 (100.0%)	10/11 (90.9%)
Gemma3:12B	Low numeric	11	11/11 (100.0%)	11/11 (100.0%)
Mistral	Example	11	10/11 (90.9%)	10/11 (90.9%)
Mistral	Word	11	10/11 (90.9%)	10/11 (90.9%)
Mistral	High numeric	11	11/11 (100.0%)	9/11 (81.8%)
Mistral	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
LLaMa-Pro	Example	11	10/11 (90.9%)	11/11 (100.0%)
LLaMa-Pro	Word	11	11/11 (100.0%)	9/11 (81.8%)
LLaMa-Pro	High numeric	11	11/11 (100.0%)	11/11 (100.0%)
LLaMa-Pro	Low numeric	11	11/11 (100.0%)	10/11 (90.9%)
Overall	All	176	173/176 (98.3%)	162/176 (92.0%)

Note. Direction stability indicates that the posterior median OR remained on the same side of one under the regularizing, primary, and weak priors. Conclusion stability indicates that the inferential classification was identical under all three priors. Rates were calculated directly from the 176 contrast-level records.

Table A12. RQ3 importance contrasts with prior-dependent inferential conclusions.

Model	Anchor type	DC solution	Regularizing	Primary	Weak
Phi-4	Word	DC connectors	2.764 [0.982, 8.945]; Uncertain	7.135 [1.515, 55.912]; Higher under Anchor	18.259 [2.161, 612.043]; Higher under Anchor
Phi-4	High numeric	Smart and sustainable DC cables	0.437 [0.140, 1.417]; Uncertain	0.180 [0.041, 0.957]; Lower under Anchor	0.104 [0.017, 0.657]; Lower under Anchor
Phi-4	Low numeric	DC connectors	2.092 [0.805, 6.264]; Uncertain	3.611 [0.956, 20.708]; Uncertain	5.046 [1.110, 51.317]; Higher under Anchor
Gemma3:12B	Example	DC-DC converter	2.366 [0.892, 6.010]; Uncertain	6.462 [1.759, 23.601]; Higher under Anchor	12.988 [2.815, 62.114]; Higher under Anchor
Gemma3:12B	Example	Protection DC system design tool	0.343 [0.102, 1.237]; Uncertain	0.124 [0.023, 0.700]; Lower under Anchor	0.073 [0.009, 0.496]; Lower under Anchor
Gemma3:12B	High numeric	DC connectors	1.636 [0.595, 4.111]; Uncertain	3.920 [0.959, 13.521]; Uncertain	7.591 [1.633, 35.109]; Higher under Anchor
Mistral	Example	DC connectors	0.426 [0.191, 1.019]; Uncertain	0.207 [0.079, 0.594]; Lower under Anchor	0.146 [0.050, 0.445]; Lower under Anchor
Mistral	Example	Protection DC system design tool	1.045 [0.466, 2.527]; Uncertain	0.913 [0.352, 2.654]; Uncertain	0.852 [0.325, 2.733]; Uncertain
Mistral	Word	DC connectors	1.325 [0.497, 4.001]; Uncertain	1.030 [0.290, 4.940]; Uncertain	0.815 [0.195, 5.322]; Uncertain
Mistral	Word	Protection DC system design tool	0.579 [0.272, 1.311]; Uncertain	0.440 [0.184, 1.097]; Uncertain	0.392 [0.164, 0.995]; Lower under Anchor
Mistral	High numeric	Semiconductor-based circuit breaker	0.436 [0.154, 1.342]; Uncertain	0.242 [0.066, 0.974]; Lower under Anchor	0.189 [0.045, 0.861]; Lower under Anchor
Mistral	High numeric	Smart and sustainable DC cables	0.437 [0.136, 1.281]; Uncertain	0.169 [0.021, 0.836]; Lower under Anchor	0.067 [0.002, 0.632]; Lower under Anchor
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Note. Cells report posterior median ORs, 95% credible intervals, and inferential conclusions. Only contrasts whose direction or conclusion differed across the regularizing, primary, and weak priors are listed.

Table A12. RQ3 importance contrasts with prior-dependent inferential conclusions.

Model	Anchor type	DC solution	Regularizing	Primary	Weak
Mistral	Low numeric	Smart and sustainable DC cables	0.437 [0.131, 1.254]; Uncertain	0.169 [0.020, 0.884]; Lower under Anchor	0.067 [0.002, 0.601]; Lower under Anchor
LLaMa-Pro	Example	LVAC-LVDC interlink converter	0.932 [0.375, 2.159]; Uncertain	1.028 [0.340, 2.686]; Uncertain	1.070 [0.337, 2.860]; Uncertain
LLaMa-Pro	Word	DC connectors	0.560 [0.241, 1.223]; Uncertain	0.431 [0.143, 1.063]; Uncertain	0.391 [0.131, 0.974]; Lower under Anchor
LLaMa-Pro	Word	DC solution design tool	0.458 [0.179, 1.058]; Uncertain	0.329 [0.096, 0.911]; Lower under Anchor	0.289 [0.073, 0.849]; Lower under Anchor
LLaMa-Pro	Low numeric	DC solution design tool	0.461 [0.184, 1.033]; Uncertain	0.332 [0.103, 0.907]; Lower under Anchor	0.286 [0.071, 0.850]; Lower under Anchor

Table A13. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Gemma3:12B	DC connectors	Word	0.029	0.919	+0.888	[0.767, 0.949]	> 0.999	Greater
Gemma3:12B	DC connectors	Example	0.029	0.010	-0.017	[-0.048, 0.035]	0.160	Uncertain
Gemma3:12B	DC measurement device	Word	0.376	0.954	+0.574	[0.456, 0.659]	> 0.999	Greater
Gemma3:12B	DC measurement device	Example	0.376	0.147	-0.228	[-0.348, -0.068]	0.004	Reduced
Gemma3:12B	DC solution design tool	Word	0.996	0.997	+0.001	[-0.022, 0.010]	0.596	Uncertain
Gemma3:12B	DC solution design tool	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.597	Uncertain
Gemma3:12B	DC-DC converter	Word	0.023	0.916	+0.891	[0.770, 0.955]	> 0.999	Greater
Gemma3:12B	DC-DC converter	Example	0.023	0.131	+0.107	[0.021, 0.256]	0.997	Greater
Gemma3:12B	LVAC-LVDC interlink converter	Word	0.004	0.002	-0.001	[-0.010, 0.026]	0.398	Uncertain
Gemma3:12B	LVAC-LVDC interlink converter	Example	0.004	0.003	-0.001	[-0.010, 0.024]	0.406	Uncertain
Gemma3:12B	Network design tool for DC solutions	Word	0.738	0.971	+0.227	[0.133, 0.305]	> 0.999	Greater
Gemma3:12B	Network design tool for DC solutions	Example	0.738	0.165	-0.570	[-0.692, -0.409]	< 0.001	Reduced
Gemma3:12B	Protection DC system design tool	Word	0.992	0.995	+0.002	[-0.029, 0.017]	0.667	Uncertain
Gemma3:12B	Protection DC system design tool	Example	0.992	0.940	-0.051	[-0.158, -0.005]	0.009	Reduced
Gemma3:12B	Semiconductor-based circuit breaker	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.602	Uncertain
Gemma3:12B	Semiconductor-based circuit breaker	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.601	Uncertain
Gemma3:12B	Smart and sustainable DC cables	Word	0.971	0.950	-0.020	[-0.124, 0.026]	0.240	Uncertain
Gemma3:12B	Smart and sustainable DC cables	Example	0.971	0.950	-0.020	[-0.124, 0.027]	0.247	Uncertain
Gemma3:12B	Solid-state circuit breaker	Word	0.996	0.998	+0.001	[-0.024, 0.011]	0.606	Uncertain
Gemma3:12B	Solid-state circuit breaker	Example	0.996	0.998	+0.001	[-0.023, 0.011]	0.608	Uncertain
Gemma3:12B	Static protection system	Word	0.996	0.997	+0.001	[-0.026, 0.010]	0.615	Uncertain
Gemma3:12B	Static protection system	Example	0.996	0.997	+0.001	[-0.025, 0.011]	0.605	Uncertain

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Table A13. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
LLaMa-Pro	DC connectors	Word	0.326	0.172	-0.153	[-0.284, 0.013]	0.033	Uncertain
LLaMa-Pro	DC connectors	Example	0.326	0.057	-0.266	[-0.356, -0.146]	< 0.001	Reduced
LLaMa-Pro	DC measurement device	Word	0.077	0.038	-0.038	[-0.094, 0.056]	0.165	Uncertain
LLaMa-Pro	DC measurement device	Example	0.077	0.038	-0.038	[-0.093, 0.056]	0.159	Uncertain
LLaMa-Pro	DC solution design tool	Word	0.269	0.109	-0.159	[-0.262, -0.016]	0.014	Reduced
LLaMa-Pro	DC solution design tool	Example	0.269	0.357	+0.089	[-0.077, 0.268]	0.848	Uncertain
LLaMa-Pro	DC-DC converter	Word	0.424	0.367	-0.055	[-0.231, 0.133]	0.271	Uncertain
LLaMa-Pro	DC-DC converter	Example	0.424	0.464	+0.041	[-0.138, 0.221]	0.669	Uncertain
LLaMa-Pro	LVAC-LVDC interlink converter	Word	0.157	0.378	+0.221	[0.056, 0.406]	0.997	Greater
LLaMa-Pro	LVAC-LVDC interlink converter	Example	0.157	0.160	+0.004	[-0.111, 0.160]	0.519	Uncertain
LLaMa-Pro	Network design tool for DC solutions	Word	0.377	0.429	+0.051	[-0.133, 0.244]	0.707	Uncertain
LLaMa-Pro	Network design tool for DC solutions	Example	0.377	0.145	-0.231	[-0.350, -0.072]	0.004	Reduced
LLaMa-Pro	Protection DC system design tool	Word	0.415	0.398	-0.016	[-0.189, 0.168]	0.432	Uncertain
LLaMa-Pro	Protection DC system design tool	Example	0.415	0.334	-0.080	[-0.243, 0.106]	0.194	Uncertain
LLaMa-Pro	Semiconductor-based circuit breaker	Word	0.688	0.575	-0.112	[-0.294, 0.065]	0.112	Uncertain
LLaMa-Pro	Semiconductor-based circuit breaker	Example	0.688	0.577	-0.109	[-0.299, 0.063]	0.112	Uncertain
LLaMa-Pro	Smart and sustainable DC cables	Word	0.541	0.469	-0.071	[-0.254, 0.122]	0.240	Uncertain
LLaMa-Pro	Smart and sustainable DC cables	Example	0.541	0.501	-0.039	[-0.226, 0.151]	0.338	Uncertain
LLaMa-Pro	Solid-state circuit breaker	Word	0.678	0.543	-0.134	[-0.322, 0.041]	0.072	Uncertain
LLaMa-Pro	Solid-state circuit breaker	Example	0.678	0.639	-0.038	[-0.224, 0.130]	0.331	Uncertain
LLaMa-Pro	Static protection system	Word	0.581	0.508	-0.073	[-0.259, 0.105]	0.213	Uncertain
LLaMa-Pro	Static protection system	Example	0.581	0.665	+0.084	[-0.106, 0.248]	0.812	Uncertain
Mistral	DC connectors	Word	0.939	0.941	+0.002	[-0.106, 0.062]	0.515	Uncertain

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Table A13. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Mistral	DC connectors	Example	0.939	0.763	-0.175	[-0.342, -0.045]	0.003	Reduced
Mistral	DC measurement device	Word	0.396	0.147	-0.247	[-0.368, -0.088]	0.002	Reduced
Mistral	DC measurement device	Example	0.396	0.238	-0.157	[-0.301, 0.022]	0.040	Uncertain
Mistral	DC solution design tool	Word	0.993	0.996	+0.002	[-0.030, 0.016]	0.649	Uncertain
Mistral	DC solution design tool	Example	0.993	0.996	+0.002	[-0.029, 0.016]	0.652	Uncertain
Mistral	DC-DC converter	Word	0.653	0.412	-0.241	[-0.418, -0.049]	0.006	Reduced
Mistral	DC-DC converter	Example	0.653	0.732	+0.079	[-0.102, 0.231]	0.807	Uncertain
Mistral	LVAC-LVDC interlink converter	Word	0.619	0.219	-0.398	[-0.543, -0.222]	< 0.001	Reduced
Mistral	LVAC-LVDC interlink converter	Example	0.619	0.316	-0.303	[-0.466, -0.116]	0.001	Reduced
Mistral	Network design tool for DC solutions	Word	0.994	0.996	+0.001	[-0.028, 0.014]	0.635	Uncertain
Mistral	Network design tool for DC solutions	Example	0.994	0.996	+0.001	[-0.026, 0.014]	0.638	Uncertain
Mistral	Protection DC system design tool	Word	0.853	0.717	-0.134	[-0.311, 0.013]	0.038	Uncertain
Mistral	Protection DC system design tool	Example	0.853	0.841	-0.012	[-0.165, 0.100]	0.431	Uncertain
Mistral	Semiconductor-based circuit breaker	Word	0.973	0.990	+0.015	[-0.035, 0.044]	0.819	Uncertain
Mistral	Semiconductor-based circuit breaker	Example	0.973	0.990	+0.015	[-0.033, 0.045]	0.826	Uncertain
Mistral	Smart and sustainable DC cables	Word	0.108	0.042	-0.064	[-0.127, 0.031]	0.073	Uncertain
Mistral	Smart and sustainable DC cables	Example	0.108	0.094	-0.013	[-0.099, 0.117]	0.406	Uncertain
Mistral	Solid-state circuit breaker	Word	0.996	0.998	+0.001	[-0.024, 0.011]	0.606	Uncertain
Mistral	Solid-state circuit breaker	Example	0.996	0.997	+0.001	[-0.026, 0.011]	0.604	Uncertain
Mistral	Static protection system	Word	0.990	0.981	-0.007	[-0.078, 0.014]	0.298	Uncertain
Mistral	Static protection system	Example	0.990	0.994	+0.004	[-0.034, 0.022]	0.693	Uncertain
Phi-4	DC connectors	Word	0.855	0.977	+0.118	[0.038, 0.184]	0.995	Greater
Phi-4	DC connectors	Example	0.855	0.748	-0.106	[-0.275, 0.032]	0.071	Uncertain

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Table A13. RQ3 importance: primary-prior upper-end probability contrasts for the word and example anchors.

Model	DC solution	Anchor	Context $P(Y = 4)$	Anchor $P(Y = 4)$	Δ	95% CrI	$P(\Delta > 0)$	Interpretation
Phi-4	DC measurement device	Word	0.556	0.910	+0.350	[0.213, 0.456]	> 0.999	Greater
Phi-4	DC measurement device	Example	0.556	0.373	-0.182	[-0.355, 0.005]	0.029	Uncertain
Phi-4	DC solution design tool	Word	0.996	0.997	+0.001	[-0.023, 0.011]	0.608	Uncertain
Phi-4	DC solution design tool	Example	0.996	0.997	+0.001	[-0.024, 0.011]	0.603	Uncertain
Phi-4	DC-DC converter	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.601	Uncertain
Phi-4	DC-DC converter	Example	0.996	0.998	+0.001	[-0.025, 0.011]	0.606	Uncertain
Phi-4	LVAC-LVDC interlink converter	Word	0.843	0.976	+0.129	[0.047, 0.196]	0.995	Greater
Phi-4	LVAC-LVDC interlink converter	Example	0.843	0.777	-0.065	[-0.231, 0.070]	0.188	Uncertain
Phi-4	Network design tool for DC solutions	Word	0.990	0.995	+0.004	[-0.033, 0.021]	0.694	Uncertain
Phi-4	Network design tool for DC solutions	Example	0.990	0.961	-0.027	[-0.121, 0.006]	0.078	Uncertain
Phi-4	Protection DC system design tool	Word	0.989	0.994	+0.004	[-0.033, 0.022]	0.707	Uncertain
Phi-4	Protection DC system design tool	Example	0.989	0.981	-0.007	[-0.078, 0.015]	0.304	Uncertain
Phi-4	Semiconductor-based circuit breaker	Word	0.996	0.997	+0.001	[-0.022, 0.010]	0.602	Uncertain
Phi-4	Semiconductor-based circuit breaker	Example	0.996	0.998	+0.001	[-0.024, 0.010]	0.597	Uncertain
Phi-4	Smart and sustainable DC cables	Word	0.987	0.994	+0.005	[-0.033, 0.025]	0.717	Uncertain
Phi-4	Smart and sustainable DC cables	Example	0.987	0.994	+0.005	[-0.033, 0.025]	0.717	Uncertain
Phi-4	Solid-state circuit breaker	Word	0.996	0.997	+0.001	[-0.025, 0.010]	0.606	Uncertain
Phi-4	Solid-state circuit breaker	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.611	Uncertain
Phi-4	Static protection system	Word	0.996	0.998	+0.001	[-0.025, 0.011]	0.605	Uncertain
Phi-4	Static protection system	Example	0.996	0.997	+0.001	[-0.023, 0.010]	0.603	Uncertain

Note. Δ is the posterior anchor-minus-Context difference in $P(Y = 4)$. Only ratings 3 and 4 were observed in the importance data; consequently, $P(Y = 1)$ was not estimated by the fitted importance model.

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